Calculus

Basic integration

8/18/2026 Lecture

Partial derivatives

Partial derivative, which implied by ∂(Del) is considered to apply derivation to single and multi-variable function.

For multi-variable function f(x_1, x_2, ..., x_i), partial derivative is its derivative with respect to one of those variables, with the others held constant.

Definition

Define that the partial derivative with respect to x_1 in multi-variable function f as

$$\frac{\partial}{\partial x_1} f(x_1, x_2, \dots, x_i) = \lim_{h \to 0} \frac{f(x_1 + h, x_2, \dots, x_i) - f(x_1, x_2, \dots, x_i)}{h}$$

In addition, we can define total differential df as

$$df = \sum_{i}^{N} \frac{\partial f}{\partial x_i} dx_i$$

Exercises 1:

Assume that the function T depends on four unknown variables: x, y, z, t.
Find partial derivative with respect to each variable, and express its total differential if the function implies T defined as below,

$$T(x, y, z, t) = x + y + z + \frac{xyz}{t}$$

Exercises 2:

Find the derivative of given implicit function with respect to x via using partial derivatives.

$$xy + x \sin xy - \frac{1}{y} = x^2$$

Anti-derivatives

Anti-derivative is differential function F whose derivatives are equal to the original function f(x). The processing of solving anti-derivatives is called “Integration”

Definition

Antiderivative F is defined with original function f as below,

$$F'(x) = f(x)$$

We can also express the anti-derivatives with “integration” symbol as below, which is looks like stretched “S” the first letter of word “Summation”.

While $f(x) = \frac{d}{dx}F(x)$,

$$f(x)dx = dF(x)$$

$$\therefore \int f(x)dx = \int dF(x) = F(x)$$

But it’s not complete definition. let’s make sure about principle of differentiation to find what’s wrong with above form… For any function of the form: a part dependent on x plus a constant,

$$\{f(x) + 1\}' = f'(x)$$

$$\{f(x) + 100\}' = f'(x)$$

$$\{f(x) + 13898269308 \ldots\}' = f'(x)$$

…

When we differentiate a function, constant becomes zero. So, when we go backward - which is integration, we know there was *some* constant there, but we don't know what it was. We need the constant C because differentiation destroys information about constants.

Definition

Antiderivative F is defined with original function f as below,

$$\int f(x)dx = F(x) + C$$

C, called “Integration constant”.

Basic skills for finding anti-derivatives

There’re several skills to find anti-derivatives of functions.

Techniques

$$\int x^n dx = \frac{1}{n+1}x^{n+1} + C \qquad \int \frac{1}{x}dx = \ln|x| + C$$

$$\int e^x dx = e^x + C \qquad \int a^x dx = \frac{a^x}{\ln a} + C$$

$$\int \sin x \; dx = -\cos x + C \qquad \int \cos x \; dx = \sin x + C$$

$$\int (\sec x)^2 \; dx = \tan x + C \qquad \int (\csc x)^2 \; dx = -\cot x + C$$

$$\int \sec x \tan x \; dx = \sec x + C \qquad \int \csc x \cot x \; dx = -\csc x + C$$

Exercises 1:

Find the anti-derivative of given function:

$$f(x) = x^2$$

Exercises 2:

Prove that the anti-derivative of e^x same as itself.

Exercises 3:

For anti-derivative F(x) of given function, F (0) = 0. Find the anti-derivative of given function.

$$f(x) = 5x^4$$

Hint) Do we have to add integration constant?

Exercises 4:

For anti-derivative F(x) of given function, F (0) = 5a. Find the anti-derivative of given function.

$$f(x) = ax^{3a}$$

Exercises 5:

For anti-derivative F(x) of given function, F (0) = 89. Find the anti-derivative of given function.

$$f(x) = 5x^4 + 4x^3 + 3x^2 + 2x + 5$$